A New Approach

Integrated Plan

PINK BOLLWORM MANAGEMENT

ALI ARSHAD

Agriculture Department
Government of Punjab

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PREAMBLE

Cotton production in Pakistan is vital for economic development of the country. We are largely dependent on the cotton industry and the related ginning and textile sector. The country witnessed record production of cotton of 15 million bales in the year 2011. But it has shown a steady decline in the following years limiting its production to 7-8 million bales per annum in the following years to follow.

One of the main reasons for this decline is the increasing attack of pests. Among these pests, pink bollworm is one of the most destructive pests of cotton crop. After getting shelter inside the cotton boll, most of the insecticides are unable to manage larvae of Pink bollworm. No pesticide spray proves effective against the larvae of the pink bollworm because of its concealed feeding inside the bolls. In previous few years, efforts were made to control it through cultural practices, however, this management was confined only to turning of heaps of cotton sticks by the limited staff of Agriculture Department in coordination with the private sector. This activity was able to move resulted in turning only 3 to 4 % of the total number of heaps of cotton sticks, leaving negligible no significant impact.

This situation direly needed needs of an out of the box solution. A comprehensive Integrated Plan for Off-Season Management of Pink Bollworm" was developed in the month of December, 2019 in consultation with Technical Advisory Committee and all other stakeholders. Besides introducing this new plan of Pink bollworm management in all cotton growing districts, only Tehsil Shuja Abad was selected for its aggressive implementation as a pilot project. This plan has ushered in a new era in the management of pink bollworm.

However, the present Plan includes a chapter on seasonal management of pink bollworm also. If implemented as conceived, it can reduce the losses of cotton crop by pink bollworm to a minimum level.

Ali Arshad Rana
Additional Secretary (Task Force)

September, 2020

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INTRODUCTION

Pink bollworm (*Pectinophora gossypiella*-Saunders) is an important pest of cotton throughout the world posing serious threat to Asia and Africa than America. The available literature suggests its origin in India which is supported by collection of some specimens from India in 1843 (Noble 1969), but recent work done by F. E. Legner at the University of California, Riverside, postulates its origin in northwest Australia and Indonesia. This pest has been found in Pakistan since the cultivation of cotton in this country, but its proper data of infestation was not available/recorded till the Establishment of Pest Survey and Forecasting in 1974.

This pest not only causes yield reduction, but it also deteriorates the lint quality by causing yellow spots in its fiber thus discounting the cost in the international market. Depending on the extent of infestation and weather conditions, this pest can cause about 20 to 30 per cent crop loss. It is estimated that the yearly loss from pink bollworm in Pakistan is about one million bales. It has inflicted heavy damages to cotton productivity in years 1965, 1983 and 2015-2019.

In Pakistan different strategies were adopted for the control of this pest in different times. Till 2001, pink bollworm was controlled by use of pesticides. During 2002 to 2010, efforts were made to control this pest by using sex pheromone and light traps. In the meantime, during 2004-2008 (4 years) and 2018 to date, PB ropes were introduced as an IPM tool to control this menace. But PB ropes were used on a small scale due to the issue of availability. Since the introduction of Bt cotton, transgenic cultivars are also used as pink bollworm management tool especially during 2009-15 (6 years). After development of resistance in cotton pink bollworm against Bt toxins, Bt varieties were used to control this pest partially in addition with chemical control from 2016 to date. In 2017, Pest Warning Department of Punjab came up with a novel idea to control pink bollworm in the months from November to April and termed this technique as off season management of pink bollworm. This practices is still in progress but during 2019-20 this off season management was augmented with imposition of Section 144 both for removal of leftover bolls from the cotton sticks and on early sowing of cotton and.

In USA, Pink bollworm has been controlled using transgenic cotton (Boll guard-3), insect pheromones to disrupt mating, releasing sterile insects to prevent reproduction, and extensive survey.

The pink bollworm is native to Asia but has become an invasive species in most of the world's cotton-growing regions. Pink bollworm is one of the most destructive pests of cotton crop. It attacks the plant squares when there are no flowers and bolls. Later on, it prefers flowers, squares and immature bolls for It lays egg laying on the squares, flowers and bolls. Soon after hatching, the first instar larva enters the fruiting bodies within 30 minutes after hatching, by making a small hole in them. Rosetted Rosette shape flowers are main identification character of this pest in case of its attack on flowers in early season. After entering the bolls, the larva closes the entry point and continues feeding inside. It starts growing there by feeding on the immature cotton seed and destroying the quality of lint with excreta. The infested boll fails to open or it appears in deformed opening opens in a very awkward/unusual fashion. At reducing the day length in winter, on lowering the temperature in winter it starts hibernating in the bolls/trashes as larvae.

After last picking in early winter, the pink bollworm infested bolls remained are left with cotton sticks due to their immature opening. These cotton sticks are removed from the fields and stored in the shape of millions of heaps lying in fields, on roadside, on in streets and inside the houses for future use as firewood. Almost every cotton stick bears left over bolls. These cotton sticks along with leftover bolls become hatcheries source of outbreak of millions of pink bollworms adults in the coming spring season.

With the rise of temperature increase of day length in early summer (March-April), it comes out of diapause, completes its pupation period and starts emerging as adult. If host plant is available, these adults give birth to their next generation by egg laying, thus multiplying their number in millions. This army of millions attacks the cotton crop, which is the only host and the destructive cycle goes on the same crop.

Due to its hiding concealed feeding inside the bolls, chemical control doesn't prove effective, hence leaving the technique of integrated pest management as the only option for its management control.

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LIFE CYCLE OF PINK BOLLWORM

In normal conditions, the entire life cycle of a pink bollworm is completed in 25-31 days. Its life cycle starts with an egg. An adult female lays about 100 to 250 eggs in a life time. Eggs are usually laid singly embedded inside the brackets of squares and under the calyx of small bolls. Eggs of pink bollworm are very small, microscopic and yellow in color. Eggs normally hatch in 3 to 7 days. Pink bollworm has four larval stages/instars i.e.

- 1st instar larva is white and takes almost 2 to 3 days to convert itself into the next instar.
- 2nd instar larva turns light pink and lasts takes for almost 4 to 5 days to convert itself into the next instar.
- 3rd instar larva becomes more pinkish and also takes 4 to 5 days to convert itself into the next instar.
- 4th instar larva turns dark pink and takes 6 to 7 days before pupation to convert itself into the next instar pupa.

At hatching, the larva has a dark head, but as it matures, the head becomes yellowish-brown in colour. It enters the fruiting bodies soon after hatching from the eggs. As it damages feeds on squares, flowers & and bolls, it grows into a length of about ¾ inches. Larva feeds for 08 to 15 days before pupationting. Larva after full fed usually leaves the boll fruit and pupates in the soil or plant debris. Pupa takes requiring about for 6-20 days for to transforming into an adult. Larva can hibernate for 90-180 days.

Pupa is reddish-brown and measures 7 to 8 mm in length. The adult pink bollworm varies in size but is rarely over 3/8 inch in length. It is mottled grey in colour with oval shaped wings. The wings are strongly fringed at the edges. The forewing is pointed at the tip. When at rest, the moth's wings are folded close to the body. Adult can live up to 10-15 days.

Mating of adults occurs after 2 to 3 days of after emergence. Pink boll worm has 4 to 6 generations in a year. Under cool dry conditions, its larvae may undergo diapause in a small cocoon in partially opened bolls in cotton lint, stored seed or in the soil. Diapausing larvae normally emerge when conditions are favourable (late March or early April) but may remain quiescent for up to 2.5 years.

It has two types of life cycles:

Short Life Cycle (suicidal)

In short life cycle, larvae pupate immediately and continue to emerge as winged adult throughout the year. Peak moth emergence occurs in March and April and most of them die due to non-availability non availability of the host and high temperature. This phenomenon is called suicidal emergence of pink bollworm.

Long Life Cycle

On start of When the winter starts, this pest overwinters and hibernates in the form of larvae in the leftover bolls. It undergoes diapause remains there by joining two seeds inside the leftover bolls present in the heaps of cotton sticks, debris and cotton seed. In July and August when the fruiting bodies are at peak, larvae come out of diapause, pupate and moths emerge within a few days. Such The larvae of this generation cause maximum damage.

Life cycle of Pink Bollworm

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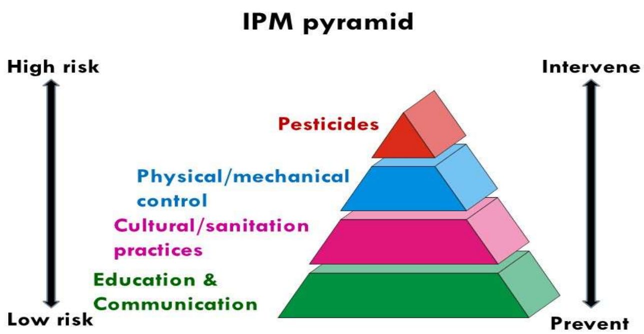
MANAGEMENT OF PINK BOLLWORM

In parts of the Asia, the pink bollworm is now becoming resistant to first generation transgenic Bt. Cotton/IR Cotton (Boll guard-I cotton) that expresses a single Bt. gene (Cry1Ac). The its breeders of locally prepared transgenic Bt. Cotton have admitted that these this varieties are is not ineffective against the pink bollworm in Pakistan sub-continent as no proper protocol were followed by farming community for sowing of Bt. Cotton (Planting ratio of Bt. and non Bt. Cotton). Infestation on susceptible cotton is generally controlled with insecticides. Un-harvested bolls harbor the larvae, so needs to be destroyed completely. If the plants cotton sticks are incorporated in the soil by rotavating rotavation rotavated in the soil and the fields are irrigated after last picking, then remaining larvae may be killed. Surviving bollworms will overwinter in the leftover bolls attached with cotton sticks and plant debris which re-infest cotton crop in the following season.

Populations of bollworms are also controlled with mating disruption, chemicals, and releases of sterile males which mate with the females but fail to fertilize their eggs. Despite all out efforts made during off season, this pest can play havoc with the cotton crop. Hence it is mandatory to have a close check on its population during the cotton season. Management of pink bollworm during cotton season can be divided in two parts i.e. monitoring and control.



Monitoring

Larvae of pink bollworm hide itself themselves in fruiting bodies, hence the monitoring through pest survey must include dissection of bolls, which itself causes damage to the crop. Hence light and pheromone traps, and rosette flowers are important tools to monitor its population of pink bollworm without damaging the crop bolls. There are three major tools for the monitoring of pink bollworm:

Light Traps

Pink bollworm belongs to the insect family Ghelechiidae and; order Lepidoptera. The adult members of this family (adults) are nocturnal in nature i.e. active during night. The light traps are an effective tool for monitoring of this pest. Three moths of this pest, trapped during 3 consecutive nights in light per trap is considered as Economic threshold Level (ETL) of pink bollworm.

Pheromone Traps

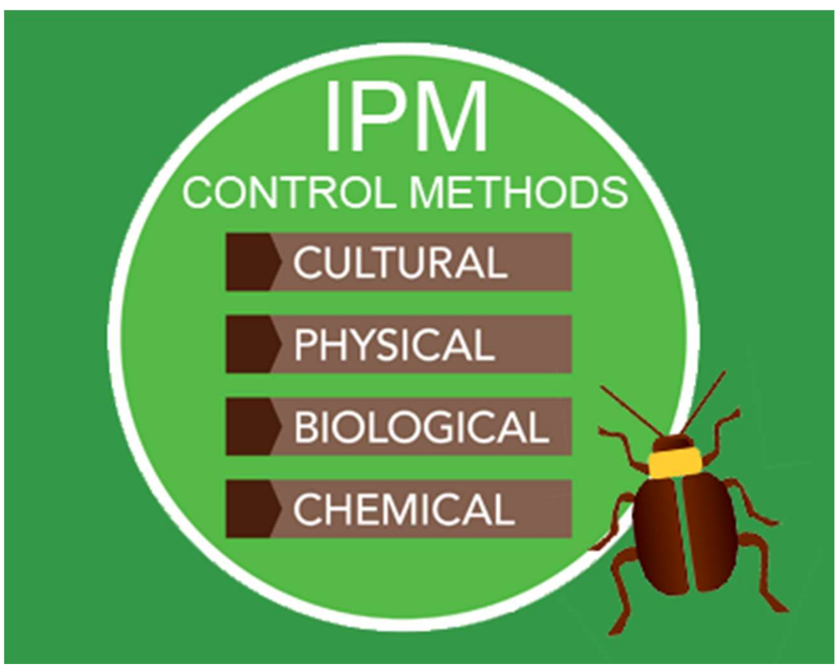
For mating, adult females of this pest secrete a pheromone called gossyp lure, which attracts the male adult. This pheromone is extracted by scientists and is now commercially available. Sex pheromone traps containing gossyp lure are effective tools for monitoring of pink bollworm adult males. Average week population of 8-9 male adults/night Three male adults of this pest, trapped during 3 consecutive nights per light pheromone trap is considered as economic thresh hold ETL of pink bollworm (Nasir M., 2019).

Pest Survey

To determine population of this pest in field, pest survey is the only tool. Different pest scouting methods are used for its survey. Up to August, 10% and upto September, 5% damage/rosette flowers is considered as its economic threshold level (ETL). This damage is ascertained by pest scouting.

Management of Pink Bollworm in Field

After careful monitoring, control of pink bollworm in the field is inevitable. Though chemical control is the last option when population of pink bollworm exceeds above economic threshold level, yet all other control measures should be continue to keep this menace at lower ebb. The measures/tactics of integrated management program of pink bollworm during the cotton season can be bifurcated categorized as;



Legislative Control

Early cotton sown in February-March can provide host to the suicidal generation of pink bollworm. Hence imposition of section 144 CrPC for ban on sowing of early cotton (before 15 April) is the only tool to prevent the farmers from early sowing of cotton. Cotton sowing date should be fixed in time frame for different ecological zones. Early sowing of cotton crop before the dates fixed by the Government should be prohibited and may be disposed-off and destroyed under the Punjab Agricultural Pests Ordinance, 1959. Sowing window for core area must be 15th April to 31th May, and for non-core area sowing date should be 15th April to 15th May. Under the Punjab Cotton Control Act 1966, the disposal of waste of ginning factories and oil mills is compulsory.

Biological Control

The biological control includes development of resistant varieties and incorporation of latest Bt genes (boll guard III); development of tolerant and early maturing varieties to escape the attack of pink bollworm and conservation of natural enemies by avoiding unnecessary application of chemicals especially pyrithroid.

Parasitoids

This documents is for local community . it will be better to mention only local predators and parasitoids reported from Pakistan.

During the past 80 years, 160 species in 43 genera of parasitic Hymenoptera have been collected with or reared from pink bollworm infested cotton. The genera *Apanteles*, *Bracon*, *Brachymeria*, *Chelonus* and *Elasmus* among the parasitic Hymenoptera contribute numerous species of potentially useful pink bollworm parasitoids.

Insecticide applications eventually eliminated attempts to biologically control pink bollworm. Insecticides represent the primary control measure which has been successful in limiting preventing damage of pink bollworm in commercial cotton. However, during more than 40 years of application, insecticides have not solved the problem of pink bollworm infestation anywhere in the world. Each growing season finds pink bollworm present and developing resistance to toxic compounds.

Predators

Green lacewing, Spiders, Coccinellids, dragon flies, wasps, damselflies, and lady beetles were the major predators of pests of cotton. All of the common predators in cotton fields are capable of feeding on pink bollworm eggs and first instar larvae. Studies show that several groups of arthropods attack pink bollworm naturally. They include mites, predaceous Dermaptera, Hemiptera, Coleoptera and Neuroptera. The egg stage is most vulnerable to attack by predators because it is relatively more exposed when as compared to larvae and pupae. Most predators lack morphological modification of the legs and mouthparts necessary to penetrate bolls to feed on pink bollworm larvae. The dermapteran *Labidura riparia* (Pallas) attacks all immature stages of pink bollworm including the pupa. However, this predator is not a dominant element in a predatory complex. Coleoptera are well represented with four species in four genera attacking pink bollworm. Most beetles focus on eggs and early instar larvae as prey. *Chrysoperla carnea* (Stephens) is the only neuropteran reported attacking pink bollworm, and seems to prefer eggs and early instar larvae. There is little data about the consumption of pink bollworm by predators under natural field conditions. If PB ropes installed in the field, then natural population of beneficial insects including *Chrysoperla carnea* can be conserved for the better management of cotton insect pests.

Cultural Control

The recommendation of cultural control during the season includes is destruction of pupae of pink bollworm during season by frequent hoeing and cultivation of soil followed by heavy irrigations. Uniform cotton planting and harvesting should be done during time frames recommended by the Agricultural Extension Services. Use of plant growth regular i.e. "Pix" (Mapiquat Chloride) for quick crop termination as the crop with late maturity suffers with heavy attack (50 -75 %) of the bolls showing the damaged locules in open bolls. Last irrigation to crop should not be after 30th September to avoid late fruiting and boll production.

Male Disruption Technique (PB Ropes)

PB Rope dispenser is a slow release system consisting of a wire-based, sealed polyethylene tube (8") filled with gossyp lure. PB ropes are hand twist-tied around the main stem of the cotton plant @ 150 - 200 dispensers / acre at pinhead square growth stage i.e. 40-45 days after sowing. This method provides a highly effective level of control of PBW for 90-120 days.

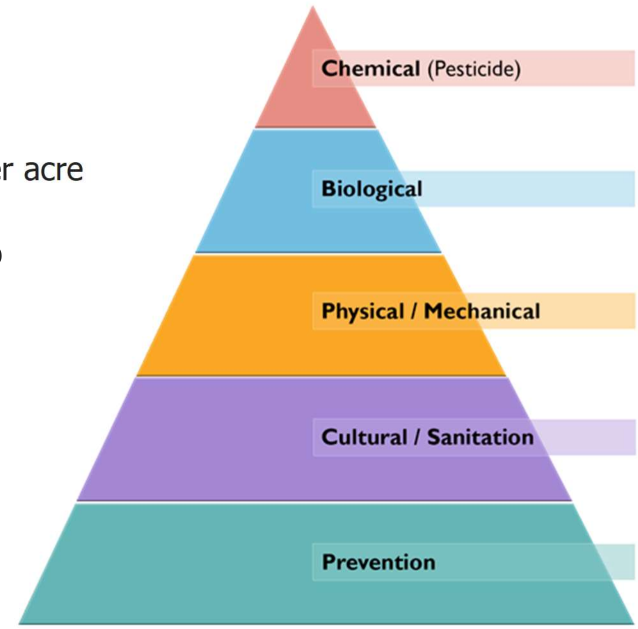
Change pic

Chemical Control

Chemical control is the last option, which should be opted very carefully after a comprehensive pest survey and only at or above economic threshold level. Novel pesticides should be preferred and hard pyrethroids should be avoided at early stage to conserve the beneficial fauna. Pesticides recommended for pink bollworm are given as:

Pesticides of Synthetic Pyrethroid group

- Bifenthrin @ 330 ml
- Lambda Cyhalothrin @ 330ml
- Deltamethrin @ 330 ml
- Gamma Cyhalothrin @ 125 ml per acre



Pesticides of Organo Phosphate group

- Triazophos @ 800 ml
- Profenophos @ 800 ml
- Chlorpyrifos @ 1000 ml per acre

New Chemistry pesticides

- Spintoram @ 100 ml
- Spinosad @ 50 ml
- Chlorantraniliprole @ 50 ml per acre.

Mixtures

- Deltamethrin + Triazophos @ 500 ml
- Cypermethrin + Profenofos @ 500 ml per acre.

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MANAGEMENT PLAN

Pink bollworm is a monophagous pest i.e. it can feed only on cotton. Cotton sowing usually starts towards the end of April every year. Management practices to control pink boll worm in its overwintering stage is called off season management. It comprises a series of management practices:

- Grazing of sheep & and goats in cotton fields after harvesting.
- Removal of leftover bolls from the standing crop.
- Post harvest rotavation of cotton sticks in soil.
- Up right stacking of heaps of cotton sticks.
- Collection of leftover bolls from heaps and its use as fuel.
- Turning of heaps after every 15 days.
- Installation of pheromone trap near heap of cotton sticks.
- Removal of ginning waste from ginning factories and oil mills and its consumption as a fuel in brick kilns before the onset of summer.
- Mass trapping of pink bollworm moths by using light and sex pheromone traps in the field.

Pink Bollworm Off-Season Management Campaign of Agriculture Department

Agriculture Department started campaign for off-season management of pink bollworm in 2017. The focus of this campaign was mainly on turning up of cotton heaps during February and March. The objective was to expose the left over bolls containing pink bollworm larvae to sunlight for early emergence of moths. Its early emergence when cotton crop is not available would cause its starvation to death. This campaign was continued in the years 2018 and 2019 but without any significant results because due to limited human and fiscal resources, it was confined to turning up of 4-5 % of the total volume of heaps.

Integrated Plan for Management of Pink Bollworm

After detailed discussions with the academia, scientists, officials of Extension Wing, officials of and Pest Warning Wing and the progressive farmers, this an integrated plan for management of pink bollworm has been prepared. Its time line is as under:

Phase-I (October to December)			
Sr.	**Activity**	**Timeline**	**Responsibility**
1	Request to Home Department for imposition of section 144 CrPC for the period of 3 months (Oct-Dec) on cutting of standing sticks after compulsory removal of bolls.	15 Sep	Additional Secretary (TF)
2	Publicity of imposition of section 144 CrPC and awareness campaign	1st Oct to 31st Dec	Director Information, All Directors, DDAs, ADAs
3	Orientation session of all officers	1st Oct to 10th Oct	Additional Secretary (TF)
4	Preparation of duty roster	1st Oct to 10th Oct	Director Information, All Directors, DDAs, ADAs (Ext.)
5	Coordination with District Administrations and lumberdars	1st Oct to 31st Dec	All DDAs, ADAs
6	Identification of cotton fields and farmers	1st Oct to 10th Oct	All ADAs, AOs
7	Serving of notices to cotton farmers	1st Oct to 31st Dec	AOs
8	Meeting with owners of brick kilns for purchase of left over bolls	1st Oct to 15th Oct	All DDAs, ADAs
9	Establishment of collection centers	1st Oct to 31st Dec	All ADAs, AOs
10	Destruction of ginning waste	1st Oct to 31st Dec	Cotton Inspectors (Ext.), ADAs, AOs PWQCP
11	Legal action against the defaulters	1st Oct to 31st Dec	All ADAs, AOs
12	Follow up	Fortnightly	All DGs, Directors, DDAs, ADAs
13	Role of PCPA, Crop Life, Cabi and NGOs	1st Oct to 31st Dec	For public private partnership and

Time line for off season management of pink bollworm in off season is given as;

Phase II (January to March)			
Sr.	**Activity**	**Timeline**	**Responsibility**
1	Request to Home Department for imposition of section 144 CrPC for the period of 3 months (Jan-Mar) for removal of leftover bolls from the heaps of cotton sticks	15 Dec	Additional Secretary (TF)
2	Publicity of imposition of section 144 CrPC and awareness campaign	1st Jan to 31st Mar	Director Information, All Directors, DDAs, ADAs
3	Orientation session of all officers	1st Jan to 10th Jan	Additional Secretary (TF)
4	Preparation of tehsil maps	1st Jan to 10th Jan	All ADAs, AOs
5	Preparation of duty roster	1st Jan to 10th Jan	Director Information, All Directors, DDAs, ADAs (Ext.)
6	Coordination with District Administrations and lumberdars	1st Jan to 31st Mar	All DDAs, ADAs
7	Identification of cotton fields and farmers	1st Jan to 10th Jan	All ADAs, AOs
8	Serving of notices to cotton farmers	1st Jan to 31st Mar	AOs
9	Meeting with owners of brick kilns for purchase of left over bolls	1st Jan to 15th Jan	All DDAs, ADAs (Ext.)
10	Establishment of collection centers	1st Jan to 31st Mar	All ADAs, AOs
11	Destruction of ginning waste	1st Jan to 31st Mar	Cotton Inspectors
12	Legal action against the defaulters	1st Jan to 31st Mar	All ADAs, AOs
13	Follow up	Fortnightly	All DGs, Directors, DDAs, ADAs

Stage-I

LEGAL

At first, it may seem ludicrous to even contemplate the use of legal measures as pest control strategy. It's easy enough to pass laws, but how do you train the insects to obey them is a big question. In reality, legal sanctions are aimed, not at the insects themselves, but at a range of human behaviors that are most likely to affect the dynamics of pest populations. Legal control tactics, therefore, include all forms of legislation and regulation that might prevent establishment or reduce spread of an insect population.

The first Quarantine Act in the US came into force in 1905, while India passed an act in 1914 titled "Destructive Insect and Pests Act of 1914" to prevent the introduction of any insect, fungus, or other pests into the country. This was later supplemented by a more comprehensive act in 1917. The legislative measures in force now in different countries can be grouped into five classes:

I. Legislation to prevent the introduction of new pests and weeds from foreign countries (international quarantine).
II. Legislation to prevent the spread of already established pests, diseases, and weeds from one part of the country to another (domestic/national quarantine).
III. Legislation to enforce farmers to apply effective control measures to prevent damage by already established pests.
IV. Legislation to prevent the adulteration and misbranding of insecticides and determine their permissible residue tolerance levels in food stuffs.
V. Legislation to regulate the activities of people engaged in pest control operations and application of hazardous insecticides.

Section 144 CrPC

Section 144 of the Code of Criminal Procedure (CrPC) empowers district administration to issue orders in public interest that may place a ban on an activity for a specific period of time. Such a ban is enforced by the police who register cases under section 188 of the Pakistan Penal Code for violations of the ban. Section 188 carries a maximum penalty of six months in prison or fine or both.

Imposition of Section 144 CrPC for control of Pink Bollworm

- Every year, in first week of September a request shall be placed by the Secretary Agriculture to the Home Department by for imposition of Section 144 CrPC that (Phase-I);
- ***"No person shall leave cotton bolls on cotton sticks before their removal from the field. No person shall retain ginning waste in ginning factories, oil mills or at any other place under his use. "***
- Every year, in first week of December a second request shall be placed by the Secretary Agriculture to the Home Department by for imposition of Section 144 CrPC that (Phase-II);

  "No person shall retain cotton bolls at fields, open places, houses, cotton ginning factories or at any other place under his use in any farm; with or without cotton sticks".
- On imposition of section 144 CrPC, a copy of the order shall be communicated to all divisional directors for onward transmission to their respective Deputy Director/Assistant Directors.
- The order shall be given a wider publicity by publication in the official Gazette, print media and electronic media as news item for guidance of the farmer.

Sr.	Activity	Responsibility
1	Website by placing the orders and related material on the main web page.	Director Information
2	Print media (All prominent newspapers) by frequent news in Agri. News Corner and by prominent advertisements at main page.	Director Information
3	Electronic media (TV channels, radio, FMs, cable networks etc) by programs and advertisement.	Director Information
4	Social media (Facebook, Whatsapp, Twitter, Instagram etc.)	Director Information, Director, Deputy Director and Assistant Director of Agri. Ext.
5	Bill boards, hoardings and burjis	Director, Deputy Director and Assistant Director of Agri. Ext.
6	Bulk SMS messages	Deputy Director of Agri. (PP), PWQCP
7	Meetings and seminars	Do
8	Face to face contact during routine activities (Pest Scouting)	All field formations
9	Farmer training programs	Director, Deputy Director and Assistant Director of Agri. Ext.
10	Use of dealers (display of flexes)	Assistant Director of Agri. (PP), PWQCP

Top Priority
Most Immediate

No. AS (TF)1-126/2019
GOVERNMENT OF THE PUNJAB
AGRICULTURE DEPARTMENT
Email: as.taskforce1@gmail.com
Dated Lahore the, December 5,.2019

The Secretary
Government of the Punjab
Home Department, Lahore

Subject: **IMPOSITION OF SECTION 144 FOR OFF-SEASON CONTROL OF COTTON PINK BOLLWORM IN COTTON GROWING DISTRICTS IN PUNJAB**

Pink Bollworm is the most threatening insect pest of cotton. Larvae of this pest hibernate in the left over bolls of the cotton. Left over bolls are the main source of hibernation of the pink bollworms larvae. Resultantly, such infested bolls become hatcheries for the insect pest which severely damage the cotton crop in the next season. The emergence of adults from the larvae of hibernating in these bolls poses huge threat to the cotton crop in terms of reduction in yield, destroying the quality of lint, besides wastage of huge resources/foreign exchange on using chemical control method. Use of chemical control increases the cost of production for farming community besides causing massive environmental pollution.

2. It has become necessary and expedient to invoke Section 144 Cr PC and to make an order to destroy the left over cotton bolls as a precautionary measure to save the cotton crop from sustaining damage from Pink Bollworm. If not circumvented now, it will not only significantly threaten the cotton production of the country but will also adversely affect the human health and environment due to excessive use of pesticides against the crop damaging insects/worm which will further lead to health issues in the population.

3. Draft Orders for imposition of Section 144 of Criminal Procedure Code, 1898 are enclosed herewith for necessary action. It is further requested to issue necessary instructions to Deputy Commissioners to implement these orders in letter and spirit following zero tolerance policy against the offenders.

Signed
(WASIF KHURSHID)
SECRETARY AGRICULTURE

CC:

1. Inspector General of Police Punjab, Lahore
2. Divisional Commissioners in Punjab
3. Director General of Agriculture (Ext & AR), Punjab, Lahore
4. Director General of Agriculture (PW & QCP), Punjab, Lahore
5. DGA (Research) AARI, Faisalabad

GOVERNMENT OF THE PUNJAB
HOME DEPARTMENT
Lahore the, 09th January, 2020

O R D E R

WHEREAS, it has been reported that Pink Bollworm is the most threatening insect pest of cotton. Larvae of this pest hibernate in the left over bolls of the cotton. Left over bolls are the main source of hibernation of the pink bollworms larvae. Resultantly, such infested bolls become hatcheries for the insect pest which severely damage the cotton crop in the next season. The emergence of adults from the larvae of hibernating in these bolls poses huge threat to the cotton crop in terms of reduction in yield, destroying the quality of lint, besides wastage of huge resources/foreign exchange on using chemical control method. Use of chemical control increases the cost of production for farming community besides causing massive environmental pollution.

WHEREAS, it has become necessary and expedient to invoke Section 144 Cr PC and to make an order to destroy the left over cotton bolls as a precautionary measure to save the cotton crop from sustaining damage from Pink Bollworm. If not circumvented now, it will not only significantly threaten the cotton production of the country but will also adversely affect the human health and environment due to excessive use of pesticides against the crop damaging insect/worm which will further lead to health issues in the population.

AND WHEREAS, in my opinion there are sufficient grounds to proceed under section 144 of Criminal Procedure Code, 1898, as an immediate prevention and speedy remedy on this issue and the direction hereinafter appearing are necessary to save the cotton crop from sustaining damage, to prevent loss to the economy of the country and to prevent public unrests in the province of Punjab.

NOW THEREFORE, I,-----Momin Agha-----, Additional Chief Secretary to Government of Punjab, Home Department, in exercise of powers vested under Section 144(6) of Criminal Procedure Code, 1898, do hereby, pass the following orders for the whole province of Punjab:

> ***No person shall retain cotton bolls at fields, open places, houses, cotton ginning factories or at any other place under his use in any form; with or without cotton sticks.***

This order shall remain in force in the Province of Punjab till **31th March, 2020** unless modified.

This order shall be given a wide publicity for publication in the official Gazette and print and electronic media as a news item.

Signed

ADDITIONAL CHIEF SECRETARY
GOVERNMENT OF THE PUNJAB
HOME DEPARTMENT

Stage-II

ORIENTATION

Orientation is a function of the mind involving awareness of three dimensions: time, place and person. Orientation is important because it lays a foundation for the employee's entire working for a new assignment in the department. First impressions are important since they establish the basis for everything that follows. Without orientation, an employee sometimes feels uncomfortable in new position and takes longer to reach full potential. Orientation is important because it:

I. Provides the employee with concise and accurate information to make him/her more comfortable in the job;
II. Encourages employee confidence and helps adapt faster to the assignment;
III. Contributes to a more effective, productive workforce;
IV. Improves employee retention; and
V. Promotes communication between the supervisor and the employee.

- Provincial orientation sessions shall be held on 1st Oct and 1st Jan every year. This session shall be chaired by the Additional Secretary, (Task Force) Punjab. All DG's of Agriculture Department shall attend this session. Director Information, Director P&E, Director CRI, Director CCRI, Vice Chancellor, University of Agri. Faisalabad, Vice Chancellor, MNS University of Agriculture Multan, all Directors of Agri. Ext. and Pest Warning of Cotton Zone, progressive growers, representatives of Crop Life and PCPA shall also attend this orientation session.
- The subsequent orientation sessions shall be held at division, district and tehsil level by respective Directors of Agri. (Ext.), Deputy Directors of Agri. (Ext.) and Assistant Directors of Agri. (Ext.) respectively within a period of seven days.
- Divisional orientation session shall be attended by all Deputy Directors of Agri. Extension and Pest Warning.

- District orientation session shall be attended by all Assistant Directors of Agri. Extension and Pest Warning.
- District orientation session shall be attended by all Agri. Officers and Field Assistants of Agri. Extension and Pest Warning.

Stage-III

AWARENESS CAMPAIGN

This measure encompasses actions that promote awareness for the altered conditions under problem and adaptation. However, not all stakeholders are aware and informed about their vulnerability and the measures they can take to pro-actively adapt to the problem. Awareness raising is therefore an important component of the adaptation process to manage the impacts of problem, enhance adaptive capacity, and reduce overall vulnerability.

I. Public awareness is important to increase enthusiasm and support, stimulate self-mobilisation and action, and mobilise local knowledge and resources.
II. Raising political awareness is important as policy makers and politicians are key actors in the policy process of adaptation.

Awareness raising requires strategies of effective communication to reach the desired outcome. The combination of these communication strategies for a targeted audience for a given period can broadly be described as 'awareness raising campaign'. The aim of awareness raising campaigns most often includes increase concern, informing the targeted audience, creating a positive image, and attempts to change their behaviour.

Raising awareness is not only important at the first stages of the process but is integral throughout the process to maintain and increase the general level of awareness. Awareness campaigns can address groups of people in a region affected by a particular threat, groups of stakeholders, the general public etc. The ultimate aim of such campaigns is to achieve long-term lasting behavioural changes. Awareness raising addresses the knowledge of individuals and organisations. It aims to ensure that all relevant regional and sub-regional bodies understand the impacts of, and take action to respond to a certain problem. However they can also focus on a certain impact that is considered as the most tremendous.

Campaigns are mostly considered as effective if several ways of communication are served: dissemination of printed materials; organisation of public meetings and training; professional consultation; communication and information through social and mass-media; using informal networks for information dissemination. It can be combined with the establishment of community self-protection teams that promote self-reliance among the farmers to minimize the risk to personal safety and property damage.

- Every year, a comprehensive awareness campaign shall be launched regarding off season management of pink bollworm. The campaign shall last from 1st Oct to 31st Dec and 1st Jan to 31st Mar for Phase-I and Phase II respectively.
- The campaign shall contain the introduction of pink boll worm, its damages, overwintering stage and social responsibility of owners of heaps, owners of ginning factories, owners of oil mills/owners of brick kilns. The consequences of non-obedience of this order shall also be highlighted in their respective areas.

- The campaign shall convey the message of the Agriculture Department as per the following plan:

Sr.	Medium	Responsibility
1	Website by placing the orders and related material on the main web page	Director Information
2	Print media (All prominent newspapers) by frequent news in Agri. News Corner and by prominent advertisements at main page	Director Information
3	Electronic media (TV channels, radio, FMs, cable networks etc) by programs and advertisement	Director Information
4	Social media (Facebook, Whatsapp, Twitter, Instagram etc.)	Director Information, All Directors, DDAs and ADAs.
5	Bill boards, hoardings and burjis	All Directors, DDAs and ADAs.
6	Bulk SMS messages	All DDAs (PW)
7	Meetings and seminars	All Directors, DDAs and ADAs.
8	Corner meetings (Pest Scouting)	All field formations
9	Farmer training programs	Director Information, All Directors, DDAs and ADAs.
10	Display of message on fertilizer and pesticide dealer shops	All ADAs and AOs

Literature and Penaflex Printing

- Wide publicity shall be made regarding imposition of Sec. 144 and awareness about off season management of Pink bollworm.
- One page pamphlets shall be printed and distributed to the farmers.
- Pakistan Crop Protection Association (PCPA) and Crop Life shall be invited and encouraged for collaboration with Agriculture Department for the awareness of Farming community and printing of literatures, penaflexes, posters, bill boards etc.

اطلاع عام

بحوالہ گورنمنٹ آف پنجاب چھٹی نمبری AS(TF),1-126/2019 مورخہ 14 جنوری 2020

ہر خاص و عام کو اطلاع دی جاتی ہے کہ

گلابی سنڈی کے تدارک کے لیے حکومت پنجاب نے دفعہ 144 کا نفاذ کر دیا ہے لہذا تمام لوگوں کو ہدایت کی جاتی ہے کہ گھروں کے اندر، باہر یا کھیت میں کپاس کی چھڑیوں پر بچے کھچے ٹینڈوں اور جننگ فیکٹریوں میں موجود کچرے جس میں گلابی سنڈی کے لاروے موجود ہیں ان کو فوری طور پر تلف کریں۔ اور آنے والی فصل کپاس کو گلابی سنڈی کے حملے سے محفوظ بنائیں۔ حکومت پنجاب کی ہدایت پر عمل نہ کرنے والوں کے خلاف قانونی کاروائی عمل میں لائی جائے گی۔ یہ حکم فوری طور پر نافذ العمل ہے۔

دفعہ 144 کی خلاف ورزی کی سزا 6 ماہ تک قید ہے

ڈپٹی ڈائریکٹر زراعت (توسیع) ملتان 061-9200748

اسسٹنٹ ڈائریکٹر زراعت (توسیع) ملتان 061-9200745

اسسٹنٹ ڈائریکٹر زراعت (توسیع) شجاع آباد 061-4396238

اسسٹنٹ ڈائریکٹر زراعت (توسیع) جلالپور پیروالا 061-4212150

گلابی سنڈی کا غیر موسمی تدارک

کپاس کی پیداوار میں کمی کا ایک سبب گلابی سنڈی ہے کپاس کے ڈھیروں سے بچے کچھے
ٹینڈے علیحدہ کر کے بھٹہ مالکان کو فروخت کریں یا تلف کریں

بروقت عمل نہ کرنے والوں کے خلاف دفعہ 144 کے تحت قانونی کاروائی عمل میں لائی جائیگی

اطلاع عام

بحوالہ گورنمنٹ آف پنجاب چٹھی نمبری AS(TF),1-126/2019
مورخہ 14 جنوری 2020

ہر خاص و عام کو اطلاع دی جاتی ہے کہ
گلابی سنڈی کے تدارک کے لیے حکومت پنجاب نے دفعہ 144
کا نفاذ کر دیا ہے لہذا تمام لوگوں کو ہدایت کی جاتی ہے کہ گھروں کے
اندر، باہر یا کھیت میں کپاس کی چھڑیوں پر بچے کھچے ٹینڈوں اور جننگ
فیکٹریوں میں موجود کچرے کو فوری طور پر تلف کریں۔ حکومت پنجاب
کی ہدایت پر عمل نہ کرنے والوں کے خلاف قانونی کاروائی عمل میں لائی
جائے گی۔ یہ حکم فوری طور پر نافذ العمل ہے۔

دفعہ 144 کی خلاف ورزی کی سزا 6 ماہ تک قید ہے

ڈپٹی ڈائریکٹر زراعت (توسیع) ملتان 061-9200748
اسسٹنٹ ڈائریکٹر زراعت (توسیع) ملتان 061-9200745
اسسٹنٹ ڈائریکٹر زراعت (توسیع) شجاع آباد 061-4396238
اسسٹنٹ ڈائریکٹر زراعت (توسیع) جلالپور پیروالا 061-4212150

کپاس کے کاشتکاروں کے نام اہم پیغام

احتیاطی تدابیر برائے گُلابی سُنڈی کا تدارک

گزشتہ چند سالوں سے ہمارے ملک میں کپاس کی مجموعی پیداوار میں بتدریج کمی کی بڑی وجہ گلابی سنڈی کے حملے میں اضافہ ہے جس کی وجہ سے ملکی معیشت کو اربوں روپے کا نقصان اٹھانا پڑ رہا ہے۔ گلابی سنڈی کے تدارک کے لیے کئے جانے والے کیمیاوی انسداد کے ساتھ ساتھ آف سیزن مینجمنٹ بھی انتہائی اہم ہے چونکہ گلابی سنڈی کپاس کا سیزن ختم ہونے کے بعد اپنی زندگی کا باقی ماندہ حصہ کپاس کی چھڑیوں کے ساتھ لگے متاثرہ ٹینڈوں میں سرمائی نیند سو کر گزارتی ہے۔ کپاس کی یہ چھڑیاں گلابی سنڈی کی اگلی فصل پر منتقلی کا سب سے بڑا ذریعہ ہیں۔ حکومت پنجاب نے گلابی سنڈی کی روک تھام کیلئے مارچ 2020ء تک پنجاب کے تمام اضلاع میں دفعہ 144 کا نفاذ کر دیا ہے۔

''جس کے تحت کوئی بھی شخص کپاس کے ٹینڈوں کو کھیتوں، کھلی جگہوں، گھروں، کپاس کے جننگ کارخانوں یا اُس کے استعمال میں کسی بھی ایسی جگہ پر کپاس کی چھڑیوں کے ساتھ یا چھڑیوں کے بغیر ذخیرہ نہیں کر سکتا''۔

تمام کاشت کار بھائیوں سے گزارش کی جاتی ہے کہ ملکی مفاد کو مدنظر رکھتے ہوئے کپاس کی چھڑیوں کے ساتھ لگے ٹینڈوں کو پہلے بطور ایندھن استعمال کر کے فوراً تلف/ جلا دیں تا کہ کپاس کو اگلے موسم میں گُلابی سُنڈی کے حملے سے بچایا جا سکے۔ خلاف ورزی کرنے کی صورت میں اندراج مقدمہ اور چھ ماہ تک قید کی سزا ہو سکتی ہے۔

منجانب پاکستان کراپ پروٹیکشن ایسوسی ایشن اینڈ محکمہ زراعت حکومت پنجاب

0800-15000, 0800-29000

CropLife
PAKISTAN

کپاس کے کاشتکاروں کے نام اہم پیغام

گزشتہ چند سالوں سے ہمارے ملک میں کپاس کی مجموعی پیداوار میں بتدریج کمی کی بڑی وجہ گلابی سنڈی کے حملے میں اضافہ ہے۔ جس کی وجہ سے ملکی معیشت کو اربوں روپے کا نقصان اٹھانا پڑ رہا ہے۔ گلابی سنڈی کے تدارک کے لیے کیئے جانے والے کیمیاوی انسداد کے ساتھ ساتھ آف سیزن مینجمنٹ بھی انتہائی اہم ہے چونکہ گلابی سنڈی کپاس کا سیزن ختم ہونے کے بعد اپنی زندگی کا باقی ماندہ حصہ کپاس کی چھڑیوں کے ساتھ لگے متاثرہ ٹینڈوں میں سرمائی نیند سو کر گزارتی ہے۔ کپاس کی یہ چھڑیاں گلابی سنڈی کی اگلی فصل پر منتقلی کا سب سے بڑا ذریعہ ہیں۔ حکومت پنجاب نے گلابی سنڈی کی روک تھام کیلئے مارچ 2020 تک پنجاب کے تمام اضلاع میں دفعہ 144 کا نفاذ کر دیا ہے۔

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منجانب۔ محکمہ زراعت حکومت پنجاب۔ **(0800-15000, 0800-29000)**

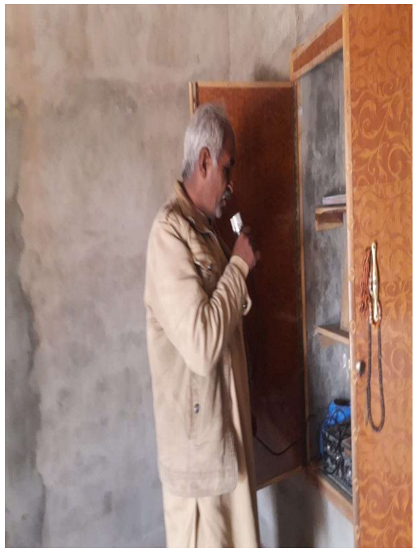
کراپ لائف پاکستان ایسوسی ایشن

لائف پلازہ نمبر 125، فرسٹ فلور، بلاک۔ ڈی ڈی، فیز 4، ڈیفنس ہاؤسنگ سوسائٹی، لاہور

فون: 19-042-35694418/ فیکس: 042-35694028/ ای میل: croplife@croplifepakistan.org

Announcements in Mosques

- Announcements shall also be made through loudspeaker in Mosques and also through loudspeaker mounted vehicles of the Agri. Department in each Mouza.



Burji

- Burjis shall be erected along the roadside carrying the messages about management of pink bollworm.

Posters

- Posters having information about pink bollworm and its management shall be pasted at prominent places.

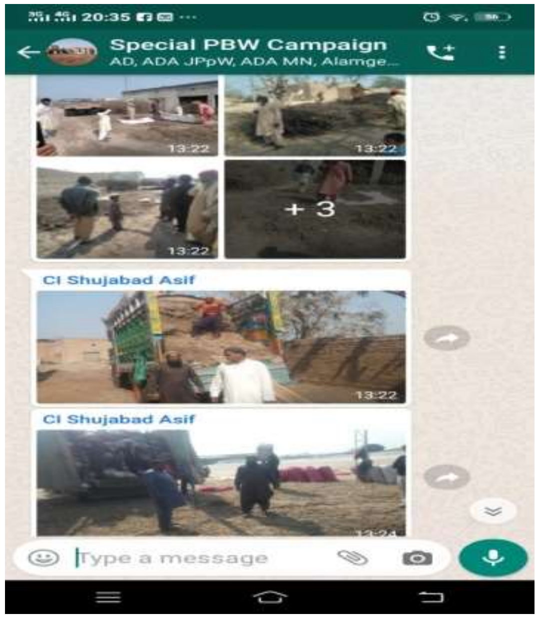
Electronic Media

- The message regarding management of pink bollworm shall be given wide coverage through news channels, interviews, talk shows and tickers.

Social Media

- Social media i.e. whatsapp group shall be used for monitoring of field staff activities during special pink bollworm campaign. Following officers/officials shall be included:
  - I. Director Agriculture Extension Multan Division Multan
  - II. Deputy Director Agriculture Extension Multan
  - III. Assistant Directors Agriculture Ext of District Multan & Shuja Abad

IV. Agriculture Officers (Ext) and Field Assistants



Stage-IV

PREPARATION

Preparation of Contingency Plan

Planning is one of the most important project management and time management techniques. Planning is preparing a sequence of action steps to achieve some specific goal. If a person does it effectively, it can reduce the time and effort for achieving the goal. A plan is like a map. When following a plan, a person can see how much they have progressed towards their goal and how far they are from their destination.

Meetings for preparation of contingency plan shall be held each year on 1st September and 1st December for phase I and II respectively.

FLOW CHART / ACTION PLAN REGARDING OFF-SEASON MANAGEMENT OF PINK BOLL WORM IN TEHSIL SHUJABAD DISTRICT MULTAN.

Tehsil Shujabad has been selected for off-season management of Pink Bollworm. It consists of 15 Union councils and 85 Mouzas/villages. Total area of Tehsil Shujabad is 286300 acres in which total cultivated area 167000 acres and area under cotton crop during year 2019 was 102000 acres. Four Supervisory Officers (Agriculture Officers) are deputed on 3 to 4 union councils each and One Field Assistant has been deputed in each Mouza/village for completion the task. The following Action is planned

<table>
<tr><th>#</th><th>Activity</th><th>Annexure</th><th>Status</th><th>Remarks</th></tr>
<tr><td>1</td><td>Flow Chart/ Action Plan</td><td>I</td><td>Completed</td><td></td></tr>
<tr><td>2</td><td>Shujabad Tehsil Map</td><td>II</td><td rowspan="2">Completed</td><td rowspan="2"></td></tr>
<tr><td>3</td><td>Deployment of Field staff & Supervisor</td><td>III</td></tr>
<tr><td>4</td><td>Detail of Brick Klins & Proposed nearest Brick klins to Field staff in each Mouza</td><td>IV</td><td rowspan="3">Will be completed in one week</td><td></td></tr>
<tr><td>5</td><td>Publicity Campaign including Penaflex, Literature and Announcements</td><td>V</td><td></td></tr>
<tr><td>6</td><td>Daily Performance report of Identification of cotton sticks heaps, preparation of list of Farmers and serving notices.</td><td>VI</td><td></td></tr>
</table>

7	Daily Performance of Establishing Collection centers	VII		
8	Daily Performance report of Collection points and sale of Cotton Bolls	VIII		
9	Development of Linkages Between Farmers and Brick klins to sell their Bolls		Will be completed in next 10 days	
10	Ensuring Collection of Cotton Bolls and their Disposal			

All this process will be completed within 17 days.

Preparation of map

Maps of all tehsils falling in the cotton zone shall be prepared bearing the following necessary information:

- Describing road network for movement and monitoring of the off-season management activities.
- Demarcation of sectors for deputing different team leaders.
- Location of UC's, mouzas and villages.
- Identification of areas with concentration of heaps of cotton sticks.
- Location of collection centres.
- Location of brick kilns.
- Location of areas with concentration of heaps.
- Ginning factories and oil mills.
- Brick kilns.
- Collection centres.
- Map printed on flex shall be displayed at notice board in offices of the Agriculture Department at Tehsil level.

نقشہ سرکل شجاع آباد
دفتر سرکل آفیسر
خانپور قاضیاں
فتح بیلہ
شیر پور
طاہر پور
جنی
موچی پورہ
سکندر آباد
UC 98
چک آرائیں
UC99
چھچھو شاہ
سومن
وینس
ماہڑہ
شجاع آباد
خوجہ
پرہاڑ
چاہان میرن خان
سانگھڑی
اگر خانی
خیر پور
رکن ہٹی
دھوندوں
غوث پور
UC 102
UC 101
بستی مٹھو
واہی نون
مقیم پور
بستی داد
نواں چک
بھانہ
شاہ موسیٰ
ماڑی نون
لطف پور
UC 105
کوٹلی نجابت
UC 104
UC 107
اوبوڑہ شمالی
لوڑی
سرائے
کیسر پور

Variable Data

This primary data shall be collected every year on changing cotton crop cultivation, areas, owners of heaps and its location. This sort of data collection shall be started right after the complete harvesting of cotton crop each year. It shall comprise cotton fields, identification of heaps, their exact locations, complete data of owners of heaps i.e. name, parentage, CNIC number, address, contact number etc.

Fixed Data

Fixed data shall include making lists of all ginning factories, oil mills, brick kilns, villages, union councils, lumberdars, collection centres and debuted field staff to be deputed. Data base shall be developed and updated on yearly basis.

LIST OF BRICK KILNS SHUJABAD

No	Name of Brick Klins and Address	Cell No
1	Noon Bro Brick Klin Jalla Pur Road Chah Muhani Wala Bangala More Shujabad	Rana Yousaf 03006309878
2	Zamri Bricks Jalal Pu Road Kanda Raheem Bux Shujabad	Hassan Khan 03007182425
3	Tameer Pakistan Bricks Company Hatti Sial Shujabad	Dilber Khan 03027786668
4	Itthad Bricks Company Sikenderabad Shujabad	Malik Bilal 03007019999
5	Khan Bhatta Company Chajju Shah	Ghulam Mustafa 03045790200
6	Noor Bhatta Company Hatti Sial	Sajjad Hussian 03027472772
7	Azeem and Bhatta Company Bangala More Shujabad	Muhammad Tufail 03013261696
8	Ghousia Mahria Bhatta Basti Mithu Shujabad	Muhammad Shafi 03017551801
9	Roshni Bricks Bhatta Company Sikenderabad	3017509819
10	New Itehad Bricks Soman	Munshi Abid 03052464063
11	Almakka Bricks Company Sikenderabad	30173004360
12	Itafaq Bhatta Company Sikender Abad Shujabad	3029005063
13	Tariq Bhatta Company Basti Dad	Tariq Khan 03007338584
14	Haji Shareef and Company Raja Ram Shujabad	Shareef 03027316600
15	Malik BROTHERS Bhatta Cmpany Jalal pur road kanda raheem bux	3457343330
16	Alraheem Bhatta Pul Dhorry wala	3016009936
17	Durani Noor Bhatta Compay	3013261696
18	Al Rehman Bhatta Sikender Abad More Shujabad	3006398822
19	Jaffer Bhatta Cmpnany near jalal pur by pass Shujabad	Gull Sher03007197239
20	Jagwal Brothers Bangala More Shujabad	Ayaz Khan 03017546544
21	Noon Brothers Kacha Pakka Shujabad	Rana Kamran 03024949909
22	Fida Hussain Bhatta Company Nawan Shahar Shujabad	Kafeel Khan 03007366216
23	Zahoor and Sons Bhatta Company	Zahoor Ahmad 03216366016
24	Inam Khan and Sons Bhatta Company	Inam Khan 03007306589

25	Magasi Brothers Rukan Hatti	Husnain Mgasi 03007182406
26	Bodla Bricks Company Shujabad	Bashir Bodla 03002564272
27	Haji Nazar Hussain Jagwal bhatta gasht	Ahmad Jagwal 03003716216
28	Aalim Bricks Company Near Jalal pur by pass	Alam Khan 03377550033
29	Almakka bhatta company near by pass (Rice Factory Shujabad)	Jalil Jagwal 03016310524
30	Maqbool Hussain Bricks Company Jalal Pur Parmat Shujabad	3017500037
31	Usman Khan Bhatta Ada Peer Ghaib Shujabad	Usman Khan 03017572576
32	Luqman Khan Bhatta Adda Peer Ghaib	3461160118
33	Khir Gull Bhatta Adda Peer Ghaib	3007320174
34	Haji Bilal Hassan Bhatta Hassan Aabd Shujabad	3036546308
35	Khwaja Bricks Company	Ayub 03015367420
36	Hamid Ullah Khan Bhatta Kotli Nijabat	Hamid Ullah3457352988
37	Rao Zulfiqar Bhatta Kotli Nijabat	Rao Zulfiqar 3039303541
38	Mirza Khan Bhatta Dharmo Wala Kotli Nijabat	3457029818
39	Lal Gull Bricks Company Wahi Swaya	3002597162
40	Rana Saeed Dosti Bhatta Company Sikenderabad	Rana Babar 03018494890
41	Haq Bahoo Bhatta Company Wahi Bhakkar	Muhammad Shahid 3003856201
42	Tanvir Hussain Bhatta Nea Rilway Phattak Shujabad	Munshi Saeed 3226731722
43	Saloo Khan bricks Co Marri Noon	Eid Khan 03006808331
44	Kafayat Ullah Bhatta Company Railwy Road Shujabad	3009730344
45	Pathanoo Wala Bhatta Old Shujabad Road Chak R.S	Arshad Shah 3006396143
46	Alnaeem Bricks Company Pull Khara	Jahan Zaib Khan 03451557990
47	PIA Bricks Parmat Shujabad	Luqman Khan 03436950152
48	A One Bricks Company Chak R.S	Arshad Ilyas

Meetings (Level 1)

During the month of October and January, every year, Deputy Director of Agri. (Ext.) and Assistant Director of Agriculture (Ext.) shall convene meetings with district and tehsil administration, Deputy Commissioners, Assistant Commissioners, District Police Officers (DPOs), Deputy Superintendents Police (DSPs), Station House Officers (SHOs), Tehsil Municipal Officers (TMOs), Land Record Officers (Patwaris), officers of local government (Chairman, Union Counsels etc.). They shall be briefed about pink bollworm management campaign, its purpose, benefits, challenges regarding their complete cooperation and confidence during execution of the campaign.

Meetings (Level 2)

In the month of every October and January, Deputy Director of Agri. (Ext.) and Assistant Director of Agriculture (Ext.) shall be responsible to convene meetings with owners/managers of all ginning factories, oil mills, brick kilns, lumberdars, and prominent/progressive farmers, representatives of Crop Life, PCPA, farmers & dealers associations and shall brief about the campaign, its purpose, benefits, challenges regarding their orientation, complete cooperation and confidence building during execution of the campaign.

Meetings (Level 3)

These meetings shall be arranged as mega meetings/seminars with farmers, dealers and owners of cotton sticks heaps. This practice shall continue from October to December for Phase-1 and from January to March for Phase-II of every year. District, Tehsil and sector in-charge i.e. Deputy Director of Agri. (Ext.), Assistant Director of Agriculture (Ext.), and Agriculture Officers shall arrange these meetings at ground level for awareness, orientation, convincing farmer community for this national cause.

Deployment of staff

This activity should be performed in the form of a mega campaign and staff of whole district should be concentrated in one tehsil at least for one week each month from October to December for Phase-1 and from January to March for Phase-II of every year.

OFFICE OF DEPUTY DIRECTOR AGRICULTURE EXTENSION MULTAN

ORDER

As per directions of Additional Secretary Task Force Agriculture Department Punjab Lahore in meeting held on 30.01.2020 at MRI, Multan. The following field staff of District Multan is hereby temporarily deputed at Mouzas/ Villages of Tehsil Shujabad for effective campaign of Offseason Management of Pink Bollworm with immediate effect till further order. They shall be responsible for identificattion of Cotton sticks heaps, removal of leftover bolls from Cotton sticks heaps, preparation of list of owners of these Heaps and serving of notices under section 144 for removal of leftover bolls. They shall also be responsibles for the identification and establishment of collection centres for purchase and sale of leftover bolls for the Brick klins in their assigned areas

#	Name of Mouza	Name of FA/ AI	Contact No	Name of Beldar
1	Raja Ram	Zahid Tanveer	0305-9637156	Shehroz Zahid
2	Basti Dad	Zafar Iqbal Chattha	0300-7318349	Ghulam Yasin
3	Khalar	Abdul Rehman	0302-3500936	Haji Abdul Majeed
4	Wahi Noon	Zafar Hussain Bazmi	0302-4946581	Ashraf Mehsi
5	Mari Noon	Jam Asgar	0301-7428645	Khadim hussain
6	wahi Bakhar	Sajjad Hussain	0302-7460836	Shoukat Hussain
7	Sikandarabad Gharbi	Rana Riaz Ahmad	0301-7438571	Ali Subtain
8	Meran Khan	Azhar Iqbal	0305-6600478	Muhammad Saleem
9	Sikandarbad Sharqi	Rana Riaz Ahmad	0301-7438571	Ali Subtain
10	Daira Pur	Rasheed Ahmad Bucha	0306-7388572	Allah Ditta
11	Chak R.S	Nazir Hussain	0306-7392761	Bashir Ahmad
12	Sanghri	Asif Munir	0307-0651962	Muhammad Yamin

#	Name of Mouza	Name of FA/ AI	Contact No	Name of Beldar
13	Jai	Muhammad Abid	0308-8200309	Sajjad Hussain JPPW
14	Khan pur Qazain	M.Qasim	0305-6576415	Naseer Bux
15	Parhar	Abdul Hafeez	0300-7078312	Muhammad Rasheed
16	Gajju Hatta Gharbi	M.Ramzan	0300-7343825	Bashir Sayal
17	Gajju Hatta Sharqi	M.Ramzan	0300-7343825	Bashir Sayal
18	Shujabad	Jam Ghulam Abbas	0301-7538835	Mushtaq Ahmad
19	Miralli Wahen	Wajahat Arshad	0306-9011725	Muhammad Wakeel
20	Mahra	Naeem Nasir	0301-7507672	Fazal Hussain
21	Vanis	Abbas Ali Anjum	0306-7350590	Waseem Shah
22	Agar Khani	Imtiaz Hussain	0302-6405955	Shoukat Hussain
23	Rukan Hatti	Naeem Nasir	0301-7507672	Fazal Hussain
24	Soman	Naeem Nasir	0301-7507672	Fazal Hussain
25	Khair Pur	Naeem Nasir	0301-7507672	Fazal Hussain
26	Punjani	Muhammad Tariq	0308-7744595	Imtiaz Hussain
27	Saray	Muhammad Saleem Akhtar	0302-7330225	M.Wakeel JPPW
28	Khara	Zulfiqar Ali Anjum	0302-5738522	Shehbaz Hussain
29	Shah Pur obha Janubi	Muhammad Jaffir	0308-8894695	Zulfiqar Ali
30	Shah Pur Obha Shumali	Ahmad Jamal Amir	0302-7340150	Ishfaq Ahmad
31	Khoja	Atif Siddique	0300-7331025	Muhammad Sadiq
32	Wahi Rakhi	Amir Bashir	0301-7556959	M.Sadiq
33	Ponta	Muhammad Rizwan	0301-7525740	Muhammad Aslam
34	Sher Pur	Abubakar Siddique	0300-6370394	Allah Ditta
35	Todar Pur	Nadeem Iqbal	0300-6397530	M.Aslam QM
36	Chajju Shah	Rana Riaz Ahmad	0301-7438571	Ali Subtain
37	Mochi Pura	Sajid Hussain	0300-9753377	M.Farooq
38	Tahir Pur	Fazal Ahmad	0305-6894638	Hafiz Ibrheem
39	Fateh Bela	Asif Islam	0306-1058175	M.Sadiq QM

#	Name of Mouza	Name of FA/ AI	Contact No	Name of Beldar
40	Basti Mithu Sharqi	Amir Anwar	0342-4770036	Pervaz
41	Basti Mithu Gharbi	M.Azhar Abbas	0301-2956255	M.Akthar Ibraheem
42	Mohan Pur Sharqi	Muhammad Amir Khan	0300-6393338	Muhammad Javaid
43	Mohan Pur Gharbi	Ghulam Abbas	0300-7343814	Safdar Ali
44	Nawan Chak	Muhammad Iqbal	0348-0129338	Ghulam Abbas
45	Naseer Pur	Amjad Ayub	0301-4428822	Muhammad Iqbal
46	Gardiaz Pur	M.Fahad	0307-7599248	M.Akthar
47	Bangala	Asif Qadri	0308-4927903	Rana Mustafa
48	Sheikh Pur Drig	Tariq Hussain	0300-6383742	Allah Ditta
49	Hilil Wajha	Shahid Rasool	0300-7370342	Sinkandar Hayat
50	Dhodhu	Muhammad Ali	0301-7481996	Muhammad Iqbal
51	Sat Burji	M.Fahad	0307-7599248	M.Akthar
52	Ghous Pur	Muhammad Ishfaq Khokhar	0303-6618520	Saddam Hussain
53	Choti Sherif	Saif Ud Din	0309-7400403	Safdar Hussain
54	Bagrain	Muhammad Akram	0300-6335528	Fiaz Hassan
55	Jhangi	Muhammad Hussain	0307-5036805	Shoukat Ali
56	Seri Shujra	Arshad Ali	0303-6985124	Muhammad Tariq
57	Aman ullah Pur	Ashfaq Hussain Hashmi	0303-8998616	M.Iqbal JPW
58	Thath Makhdoom Pur	Ashfaq Hussain Hashmi	0303-8998616	M.Iqbal JPW
59	Jalal Pur Khakhi	M.Kashif Nawaz	0301-2408901	M.Tariq Javid
60	Bhana	Azhar Abbas Shah	0306-6966539	Jam Amjad
61	Punjani khas	M.Hussain Meo	0301-4803341	Bashir Ahmad Bhatti
62	Tror Jalal Pur	M.Hussain Meo	0301-4803341	Bashir Ahmad Bhatti
63	Thath Ghalwan Shumali	Tahir Masood	0304-0873450	Ghulam Qadir
64	Thath Ghalwan Janubi	Tahir Masood	0304-0873450	Ghulam Qadir
65	Dera Arbi	Adil Hussain Shah	0300-6589483	Abdul Basit

#	Name of Mouza	Name of FA/ AI	Contact No	Name of Beldar
66	Obra Shumali	Yasir Arafat	0301-8789877	Shoukat Abbas JPPW
67	Gawain	Muhammad Rafiq	0301-7558931	Sajid JPPW
68	Lasoori	M.Nabeel	0308-4256516	Khurshid JPPW
69	Jangal Ameer Hussain	Atta ul Rehman	0301-7576670	Sabir Hussain JPPW
70	Matotali	Tahir Majeed	0302-9764662	Bashir Ahmad Arain
71	Qasir Pur	Tahir Majeed	0302-9764662	Bashir Ahmad Arain
72	Voni	Tahir Majeed	0302-9764662	Bashir Ahmad Arain
73	Kalch Pur	Rashid Rehamani	0301-8781170	M.Arshad
74	Sheikh Pur Shujra	Tahir Majeed	0302-9764662	Bashir Ahmad Arain
75	Rasool Pur Shumali	Sajjad Hussain	0308-5133917	Abdur Rehman
76	Rasool Pur Junubi	Sajjad Hussain	0308-5133917	Abdur Rehman
77	Maqeem Pur	Aslam PS PP SJB	0302-7332799	Beldar PP
78	Shah Mousa	Aslam PS PP SJB	0302-7332799	Beldar PP
79	Lutaf Pur	Aqeel PS PP	0302-7486543	Beldar PP
80	Ghabi Jakhar	Aqeel PS PP	0302-7486543	Beldar PP
81	Kotli Nijabat Janubi	Muhammad Bota	0345-7334975	Abdul Rehman
82	Kotli Nijabat Shumali	Muhammad Bota	0345-7334975	Abdul Rehman
83	Chak Jakhar	Muhammad Akbar	0302-7419036	Sajjad Hussain Shah
84	Chak 13 Fiaz	Rana Nisar	0300-4984433	Shoukat Hussain
85	Wahi Sawaya	Rana Nisar	0300-4984433	Shoukat Hussain

Stage-V

EXECUTION

As Steve Jobs once said, “To me, ideas are worth nothing unless executed ... Execution is worth millions.” Even if you brainstorm a brilliant strategy, it will be wasted if it's never fully realized. So, while other leaders are focused disproportionately on idea-generation, consider these three suggestions for gaining competitive advantage.

Execution of plan is vital to an organization's success. An organization must be able to efficiently execute that strategy to achieve its performance improvement goals. The organization's culture is often the most important determiner in successful execution. Prioritizing execution requires management to think of the project plan strategy a living, evolving entity, meaning routinely revisit, reanalyze and make updates based on how things are going in real-time. It enables the management to course-correct more frequently, and with better results.

- After issuance of 1st order for imposing section 144 CrPC for a period of 3 months from 1st October to 31st December, the execution stage shall be initiated. Cotton farmers shall not be allowed to cut sticks till the removal of unopened bolls.
- Similarly, after issuance of 2nd order imposing section 144 CrPC, there shall be ban on keeping heaps of cotton sticks with leftover bolls for a period of 3 months from 1st January to 31st March every year.
- One month time shall be given to the farmers and heap owners for complying with the orders of the Government.

Legal Notice and Action

- Legal notices shall be served;
  - From October to December (Phase-1) every year, to those farmers who do not comply with the orders of the department and cut the cotton sticks without removing leftover bolls.
  - From January to March (Phase-II) of every year, to the owners of heaps of cotton sticks, who do not destroy the leftover bolls.
- In case violator/accused is reluctant to receive the notice, it shall be affixed on gate/wall of his residence.
- If the farmer/heap owner does not comply with the directions of the Government despite serving of the legal notice, FIR shall be registered with in the concerned police station under section 188 PPC.

نوٹس زیر دفعہ 144 برائے تلفی کپاس کی چھڑیوں سے بچے کھچے ٹینڈے

نام ---------------- ولدیت ---------------- رابطہ نمبر ------------------------

موضع / گاؤں ---------------- یونین کونسل ---------------- تعداد ڈھیر ------------------------

تحصیل ---------------- تاریخ وصولی نوٹس ---------------- دستخط وصول کنندہ ------------------------

نوٹس زیر دفعہ 144 برائے تلفی کپاس کی چھڑیوں سے بچے کھچے ٹینڈے

بنام مکرمی و محترم جناب ---------------- ولدیت ------------------------

موضع ---------------- یونین کونسل ---------------- تحصیل ------------------------

السلام علیکم

آپکو بذریعہ نوٹس ہذا مطلع کیا جاتا ہے کہ حکومت پنجاب نے قومی اور عوامی مفاد کو مدنظر رکھتے ہوئے کپاس کی گلابی سنڈی اور ٹینڈوں کی تلفی کو یقینی بنانے کے لئے دفعہ **144** کا نفاذ کیا ہے لہذا آپکو خبردار کیا جاتا ہے کہ آپکی ملکیت اور زیر استعمال کپاس کی چھڑیوں کے ڈھیروں میں موجود ٹینڈوں میں گلابی سنڈی کے لاروے موجود ہیں جو کہ آئندہ فصل کپاس کے لئے نقصان کا باعث بنیں گے جس سے نہ صرف پھٹی کی کوالٹی بلکہ پیداوار میں بھی نمایاں کمی ہوگی۔

لہذا آپکو بذریعہ نوٹس ہذا ہدایت کی جاتی ہے کہ نوٹس کی وصولی کے بعد اندر معیاد (7) یوم کپاس کی چھڑیوں کے ڈھیروں سے بچے کھچے ٹینڈے علیحدہ کر کے تلف کریں۔ بصورت دیگر دفعہ **144** کی خلاف ورزی پر حسب ضابطہ کاروائی عمل میں لائی جائے گی۔

نام جاری کردہ ---------------- عہدہ ------------------------

مورخہ ---------------- دستخط ------------------------

محکمہ زراعت (توسیع) تحصیل ---------------- ضلع ملتان

Establishment of Collection Centres and Transportation

- Collection centres shall be established every year, from 1^{st} November to 31^{st} December and from 1^{st} February to 31^{st} March at appropriate places with the consultation of Lumberdars and farming community of the area.
- All the debris, leftover bolls, ginning & oil mills waste of a sector shall be collected at these collection centres.
- Nearest brick kilns shall be attached with these collection center for purchase of bolls.
- Tehsil in-charges shall facilitate transportation of waste from collection centres to brick kilns.

Stage-VI

REPORTING AND FOLLOW UP

Data reporting is a very important factor in the day to day activities. Data reporting goes hand in hand with data analysis and is essential in every walk of life. A proficient reporting is a part of a management control system that provides information. This information can be in the form of reports and/or statements. The system is designed to assist members of the management by providing timely pertinent information. Management reporting systems help in capturing data that is needed by managers to run an effective business. Here are the reasons why an organisation needs an effective management reporting system:

i. Constant need of reports for decision making and analysis of trends.
ii. To ensure availability of reports with the right stakeholders at the right time.
iii. For a single holistic view of the performance.
iv. To avoid data redundancy, duplication of data leading to data management and quality issues leading to error prone reports.

An effective management reporting system helps:

i. Improve decision making.
ii. Improves management effectiveness.
iii. Improves responsiveness to issues.
iv. Improve efficiency of resources in the delivery of organizational services.

A regular follow up always gives stakeholders a chance to be heard and engage effectively. Follow-ups can be a great source to ask stakeholders, "What they want/expect next."

- During all stages of the campaign, daily reports shall be generated by all concerned as per the proforma given below.

PROFORMA-A

DAILY REPORT ABOUT IDENTIFICATION OF HEAPS & NOTICES SERVED					
Sr.	Name of Mouza	Name of Farmer with Address & Cell No.	No. of Heaps	Notice Serve	Dated

PROFORMA-B

SUMMARY OF DAILY PROGRESS REPORT BY SUPERVISORY OFFICER				
Sr.	Name of Supervisor Officer	No. of Farmers	No. of Heaps Identified	No. of Notice Served

PROFORMA-C

PROGRESS REGARDING ESTABLISHMENT OF COLLECTION CENTRES			
Sr.	Name of Union council	Name of Mouza	Detail of collection centre with address

PROFORMA-D

DAILY PROGRESS REPORT OF COLLECTION CENTRES				
Sr.	Name of Village	No. of Collection Centre Established	Left Over Bolls Collected (Kg)	Left Over Bolls Purchase By Brick Klins (Kg)

- These reports shall be sharded in the PBW Management Group to be created for this purpose.

Stage-VII:

MONITORING AND EVALUATION (M&E)

Monitoring and Evaluation (M&E) is used to assess the performance of projects, institutions and programmes set up by governments, international organisations and NGOs. Its goal is to improve current and future management of outputs, outcomes and impact. Monitoring is a continuous assessment of programmes based on early detailed information on the progress or delay of the ongoing assessed activities. An evaluation is an examination concerning the relevance, effectiveness, efficiency and impact of activities in the light of specified objectives.

- Additional Secretary (Task Force) shall monitor this campaign in Punjab.
- Director General, Agriculture (Ext. & AR) shall be overall in-charge of the campaign. in close coordination with
- The Director General, Pest Warning shall be the Coordinator of the campaign.
- Directors, Deputy Directors, Assistant Directors of Agriculture Extension department shall be divisional, district and tehsil supervisors respectively and shall supervise the campaign.
- For counterchecking and monitoring of this campaign, the cotton growing divisions shall be allocated to the senior officers of the Agriculture Department as per the following details:

Sr.	Area	Responsibility
1	Overall Punjab	Additional Secretary (TF), Punjab
2	Bahawalpur Division	Director General, Pest Warning & QCP, Punjab

3	Dera Ghazi Khan Division	Director, Cotton Research Institute, Multan
4	Multan Division	Director General, Extension and AR, Punjab
5	Sahiwal Division	Director General, Water Management, Punjab
6	Faisalabad Division	Director General, Research, Punjab
7	Sargodha Division	Director General, Research, Punjab

Stage-VIII

RESULTS & DISCUSSION

The purpose of a Results section is to present the key results of this campaign. Results and discussions can either be combined into one section or organized as separate sections depending on the requirements. The results of this campaign collected through daily, weekly, monthly and annual reports shall be complied, analyzed, evaluated by Agriculture Statistics department each year till April. Agri information department shall be held responsible for publication of the results and distribution amongst all related wings, private sector and mainstream media. The results and discussion sections are one of the challenging sections to write. It is important to plan this section carefully as it may contain a large amount of data that needs to be presented in a clear and concise fashion. Results shall be very clear and shall contain subsections, subheadings and descriptive pictures of graphs/field activities to improve readability and clarity. Discussion shall also include vertical and horizontal results highlighting potential limitations. This often makes up for a great discussion section, so shall highlight the negative results also. This shall make a pile of criticism; though will definitely improve this campaign in years to come.